Differential Equations
Midterm – Fall 2022

Time: 15:20-16:35

Name:
Student ID:

1) Solve the initial value problem:

$$y' + (tanx)y = 4x^3 cosx,\ y(0) = 8.$$

(25 Pts)

2) **a)** Prove that every separable first-order differential equation is exact.

b) When is a linear first-order differential equation $(y' + p(x)y = q(x))$ separable?

(25 Pts)

3) Find the general solution of the differential equation:

$$(1 + y')cos(x + y) = yy'$$

(25 Pts)

4) Find the general and singular solution of the differential equation:

$$y = 2xy' + e^{y'}.$$

(25 Pts)

5) Find the general solution of the differential equation:

$$xsin\theta d\theta + (x^3 - 2x^2 cos\theta + cos\theta)dx = 0.$$

(25 Pts)

SOLVE ONLY 4 QUESTIONS!

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